

Abasyn University Islamabad Campus

Course Specification Form

1. Department	2. Program
Computing	BS Artificial Intelligence

3. Course Code	4. Course Title	5. Lectures Timing & Location	
AI 216	Programming for Artificial Intelligence	Monday	08:30 – 10:30
6. Credits	7. Pre-requisites	8. Course Web-Page	9. Course Coordinator
2+1	-	Lms.abasynisb.edu.pk	

10. Academic Year	11. Semester
2026-2027	Fall 2026

12. Course Instructor	13. Email	14. Office	15. Office Hours
Asad Hanif	asad.hanif@abasynisb.edu.pk		

16. Textbook(s)
1. Severance, C.R., 2016. "Python for everybody: Exploring data using Python 3." CreateSpace Independent Publ Platform. 2. Miller, B.N., Ranum, D.L. and Anderson, J., 2019. "Python programming in context." Jones & Bartlett Pub. 3. McKinney, W., 2012. "Python for data analysis: Data wrangling with Pandas, NumPy, and IPython." O'Reilly Media, Inc.

17. Reference(s)
1. Russell, S. and Norvig, P. "Artificial Intelligence. A Modern Approach", 3rd ed, Prentice Hall, Inc., 2015. 2. Joshi, P., 2017. "Artificial intelligence with python." Packt Publishing Ltd. 3. Janert, P.K., 2010. "Data analysis with open source tools: a hands-on guide for programmers and data scientists." O'Reilly Media, Inc.

18. Other Resources used (e.g. e-learning, field visits, periodicals, software, etc.):
https://www.w3schools.com/python/python_intro.asp https://pll.harvard.edu/course/cs50s-introduction-artificial-intelligence-python https://www.coursera.org/learn/ai-python-for-beginners NumPy documentation Pandas documentation scikit-learn user guide FastAPI documentation Docker documentation pytest documentation LLM provider documentation

19. Course Description (as approved by the dept.)
AI-216 Programming for Artificial Intelligence introduces students to Python programming and professional software development practices required for building practical artificial intelligence applications. The course develops proficiency in Python fundamentals, functions, object-oriented programming, data structures, debugging, error handling, clean code, modular programming, and version control using Git and GitHub. Students learn to acquire, process, analyze, and visualize data using libraries such as NumPy and Pandas and to construct reproducible machine-learning workflows using scikit-learn.

The course further introduces AI engineering practices including project organization, dependency management, testing, logging, configuration, model serialization, inference pipelines, REST APIs, containerization, and basic deployment concepts. Students are also introduced to contemporary AI application development using foundation-model APIs, structured outputs, embeddings, and retrieval-augmented generation. Emphasis is placed on converting experimental code into maintainable, reproducible, and usable AI applications through hands-on laboratories and an end-to-end semester project.

20. Course Learning Outcomes (CLOs)			
CLOs	Mapping to PLOs		
On successful completion of the course, students will be able to:	Domain	Taxonomy Level	PLO
1) Apply Python programming constructs, functions, data structures, modules, object-oriented programming, and professional coding practices to solve AI-related programming problems.	Cognitive	3	
2) Acquire, preprocess, analyze, and visualize data using Python libraries and data sources such as files, APIs, web sources, and databases.	Cognitive	3	
3) Develop and evaluate reproducible machine-learning workflows using standard Python libraries, including preprocessing, model training, validation, and inference.	Cognitive	3	
4) Develop a maintainable AI application by integrating appropriate software engineering, model-serving, reproducibility, and modern AI application-development practices.	Cognitive	4	

21. Course Assessment		
Assessment Type	Weight	Date(s)
Midterm	25	To be announced by examination
Quizzes	10	2 Before Midterm, 2 Before Final Exam
Assignments	10	Due dates as mentioned on the assignments
Final Exam	50	To be announced by examination
Project	05	Before Final Exam
Total	100	

22. Course Teaching Methods	
1) Interactive learning	
2) Problem solving learning	
3) Practical and laboratory learning	
4) Self-directed learning	
5) Group work and project	

23. Course Weekly Breakdown			
Week	Date	Topics Covered	Lab Activities
1.		Course orientation; What does an AI Engineer do? Python/AI engineering roadmap; Git, GitHub, repositories, commits	Git/GitHub setup, clone/fork, first repository, meaningful commits, README

2.		Python fundamentals: variables, operators, conditions, loops, functions	Problem-solving exercises using Python
3.		Functions, modules, exceptions, debugging, OOP for AI applications	Refactor procedural code into functions/classes/modules
4.		Python data structures; comprehensions; file organization; clean code	Data-oriented programming exercises
5.		NumPy, arrays, vectorization, numerical computing	Replace Python-loop computations with NumPy
6.		Pandas and data manipulation	Load, filter, transform and summarize a real dataset
7.		Data ingestion: CSV/JSON, APIs, web scraping, SQLite/SQL	Build a small ingest → clean → store pipeline
8.		Data preprocessing, EDA, missing values, categorical data, visualization, reproducibility	EDA and preprocessing case study
9.		Midterm Examination	
10.		Machine-learning workflow; baselines; train/test split; scikit-learn estimators and Pipelines	Build an end-to-end classification/regression pipeline
11.		Evaluation, metrics, overfitting, data leakage, cross-validation, feature engineering/selection	Compare models correctly using cross-validation
12.		AI Software Engineering: project structure, virtual environments, dependencies, configuration, logging, type hints, pytest	Convert a notebook into a structured Python project
13.		Model inference and APIs: model serialization, prediction services, HTTP, REST, JSON, FastAPI, Pydantic	Implement /health and /predict endpoints
14.		Containers and reproducibility: Docker images, Dockerfile, dependency isolation; introduction to CI	Containerize the FastAPI model service; optional GitHub Actions test workflow
15.		LLM application engineering: model APIs, prompts, system/user messages, structured output, environment variables, latency/cost/reliability	Build a small Python application using a foundation-model API
16.		Embeddings and RAG: chunking, embeddings, similarity search, retrieval, grounding, basic evaluation	Build a small “Ask My Documents” system
17.		End-to-end AI system integration; responsible use of AI coding tools; project demos and engineering review	Teams demonstrate working system and defend engineering decisions
18.		Final Examination	

Prepared by:

Date:

Approved by the Department on: